
Math 21

Algebra and its Functions

Math 21 Course at a Glance

Algebra and its Functions

TEXTBOOK | OpenStax Intermediate Algebra 2e (Marecek and Mathis, 2020)

Review Content		Functions and Function Interpretation		Polynomials and Factoring		Rational Expressions and Equations	
REVIEW	SECTIONS 3.1, 3.2, 3.3, 5.2	REQUIRED	SECTIONS 3.5, 3.6	REQUIRED	SECTIONS 5.1, 5.3, 5.4, 6.1, 6.2, 6.3	REQUIRED	SECTIONS 7.1, 7.2, 7.3, 7.4, 7.5
LEARNING OBJECTIVES		LEARNING OBJECTIVES		LEARNING OBJECTIVES		LEARNING OBJECTIVES	
M21-REV-001	I can determine and interpret the slope of a line.	M21-FUN-001	I can evaluate a function from a formula, graph, or table.	M21-POL-001	I can add and subtract polynomials.	M21-RAT-001	I can state excluded values for a rational expression.
M21-REV-002	I can graph a line from an equation and write an equation from given information.	M21-FUN-002	I can determine whether a relation is a function.	M21-POL-002	I can multiply polynomials.	M21-RAT-002	I can simplify a rational expression while preserving its restrictions.
M21-REV-003	I can apply a linear function to a context.	M21-FUN-003	I can determine domain and range from a graph or table.	M21-POL-003	I can factor out a greatest common factor.	M21-RAT-003	I can multiply and divide rational expressions.
M21-REV-004	I can identify the solution of a linear system as an intersection point.	M21-FUN-004	I can solve for an input that produces a specified output.	M21-POL-004	I can factor a quadratic trinomial.	M21-RAT-004	I can add and subtract rational expressions using a common denominator.
M21-REV-005	I can solve a simple two-variable linear system using an appropriate method.	M21-FUN-005	I can interpret inputs, outputs, domain, and range in context.	M21-POL-005	I can factor a difference of squares and other required special cases.	M21-RAT-005	I can solve a rational equation and reject extraneous solutions.
M21-REV-006	I can simplify an expression using basic positive-integer exponent rules.	M21-FUN-006	I can calculate and interpret average rate of change.	M21-POL-006	I can use factoring to determine the zeros of a polynomial.		
		M21-FUN-007	I can compare functions represented by formulas, graphs, or tables.				

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Exponents and Radicals

REQUIRED SECTIONS 8.1, 8.2, 8.3, 8.4

LEARNING OBJECTIVES

M21-EXP-001	I can simplify expressions containing negative exponents.
M21-EXP-002	I can rewrite expressions between radical and rational-exponent forms.
M21-EXP-003	I can simplify radical expressions.
M21-EXP-004	I can apply exponent rules to expressions with integer and rational exponents.
M21-EXP-005	I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED SECTIONS 9.1, 9.2, 9.3, 9.4, 9.5, 9.6, 9.7

LEARNING OBJECTIVES

M21-QUAD-001	I can identify key features of a quadratic function from its graph or equation.
M21-QUAD-002	I can connect standard, factored, and vertex forms to features of a parabola.
M21-QUAD-003	I can solve a quadratic equation by factoring.
M21-QUAD-004	I can solve a quadratic equation using the square-root property.
M21-QUAD-005	I can solve a quadratic equation by completing the square.
M21-QUAD-006	I can solve a quadratic equation using the quadratic formula.
M21-QUAD-007	I can choose an efficient method for solving a quadratic equation.
M21-QUAD-008	I can graph a quadratic function using its vertex, intercepts, and symmetry.
M21-QUAD-009	I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION SECTIONS 9.3

LEARNING OBJECTIVES

M21-EXT-001	I can perform basic arithmetic with complex numbers.
M21-EXT-002	I can express the nonreal solutions of a quadratic equation using complex numbers.

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Review Content

REVIEW

UNIT TEXTBOOK COVERAGE

3.1 Graph Linear Equations in Two Variables; 3.2 Slope of a Line; 3.3 Find the Equation of a Line; 5.2 Properties of Exponents and Scientific Notation

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-REV-001	I can determine and interpret the slope of a line. TEXTBOOK SECTIONS 3.2 Slope of a Line
SUPPORTING SKILLS	
R	Determine slope from a graph. (first introduced in Math 12) SK-LIN-SLOPE-GRAPH · Reinforce
R	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Reinforce
R	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Reinforce

M21-REV-002	I can graph a line from an equation and write an equation from given information. TEXTBOOK SECTIONS 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line
SUPPORTING SKILLS	
R	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Reinforce
R	Use slope to generate additional points on a line. (first introduced in Math 12) SK-LIN-GRAPH-SLOPE · Reinforce
R	Plot an intercept accurately on a coordinate plane. (first introduced in Math 12) SK-LIN-PLOT-INTERCEPT · Reinforce
A	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Apply

M21-REV-003	I can apply a linear function to a context. TEXTBOOK SECTIONS 3.3 Find the Equation of a Line
SUPPORTING SKILLS	
I	Explain a model parameter in the language of its context. SK-MODEL-PARAMETER-INTERPRET · Introduce
R	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Reinforce
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M21-REV-004	I can identify the solution of a linear system as an intersection point. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
R	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Reinforce

M21-REV-005	I can solve a simple two-variable linear system using an appropriate method. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
R	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Reinforce
R	Add equations to eliminate a variable. (first introduced in Math 12) SK-SYS-ELIMINATE · Reinforce
R	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Reinforce
R	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Reinforce

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M21-REV-006	I can simplify an expression using basic positive-integer exponent rules. TEXTBOOK SECTIONS5.2 Properties of Exponents and Scientific Notation
SUPPORTING SKILLS	
R	Apply the power rule for exponents. (first introduced in Math 12) SK-EXP-POWER · Reinforce
R	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Reinforce
R	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Reinforce

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SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Functions and Function Interpretation

REQUIRED

UNIT TEXTBOOK COVERAGE

3.5 Relations and Functions; 3.6 Graphs of Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-FUN-001	I can evaluate a function from a formula, graph, or table. TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions
SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Identify the input and output represented by function notation. (first introduced in Math 12) SK-FUNC-NOTATION-READ · Apply

M21-FUN-002	I can determine whether a relation is a function. TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions
SUPPORTING SKILLS	
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
D	Apply the vertical line test to a graph. (first introduced in Math 12) SK-FUNC-VERTICAL-LINE · Deepen

M21-FUN-003	I can determine domain and range from a graph or table. TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions
SUPPORTING SKILLS	
D	Read domain and range from a graph. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-GRAPH · Deepen
D	Read domain and range from a table. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-TABLE · Deepen

M21-FUN-004	I can solve for an input that produces a specified output. TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Solve for an input associated with a specified output. SK-FUNC-SOLVE-INPUT · Introduce

M21-FUN-005	I can interpret inputs, outputs, domain, and range in context. TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions
SUPPORTING SKILLS	
D	Restrict domain to values meaningful in a context. (first introduced in Math 12) SK-FUNC-CONTEXT-DOMAIN · Deepen
D	Interpret the range of a function in a context. (first introduced in Math 12) SK-FUNC-CONTEXT-RANGE · Deepen
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
D	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Deepen

M21-FUN-006	I can calculate and interpret average rate of change. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
I	Calculate average rate of change over an interval. SK-FUNC-AROC · Introduce
I	Interpret average rate of change with contextual units. SK-FUNC-AROC-INTERPRET · Introduce
D	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Deepen

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M21-FUN-007	I can compare functions represented by formulas, graphs, or tables. TEXTBOOK SECTIONSNo mapped textbook section
SUPPORTING SKILLS	
I	Compare functions shown in different representations. SK-FUNC-COMPARE-REP · Introduce
I	Connect parameters in a formula to features of its graph. SK-REP-FORMULA-GRAPH · Introduce
I	Use tabular patterns to predict graph shape and behavior. SK-REP-TABLE-GRAPH · Introduce

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Polynomials and Factoring

REQUIRED

UNIT TEXTBOOK COVERAGE

5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-POL-001	I can add and subtract polynomials. TEXTBOOK SECTIONS 5.1 Add and Subtract Polynomials; 5.4 Dividing Polynomials
SUPPORTING SKILLS	
A	Identify and combine like terms. (inherited from middle school) SK-ALG-LIKE-TERMS · Apply
D	Identify like polynomial terms by variable part and exponent. (first introduced in Math 12) SK-POLY-LIKE-TERMS · Deepen
R	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Reinforce

M21-POL-002	I can multiply polynomials. TEXTBOOK SECTIONS 5.3 Multiply Polynomials; 5.4 Dividing Polynomials
SUPPORTING SKILLS	
A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
D	Multiply two binomials using distribution. (first introduced in Math 12) SK-POLY-BINOMIAL-MULTIPLY · Deepen
D	Multiply a monomial by a polynomial. (first introduced in Math 12) SK-POLY-DISTRIBUTE · Deepen
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M21-POL-003	I can factor out a greatest common factor. TEXTBOOK SECTIONS 6.1 Greatest Common Factor and Factor by Grouping
SUPPORTING SKILLS	
R	Verify a factorization by multiplication. (first introduced in Math 12) SK-FACTOR-CHECK · Reinforce
D	Factor the greatest common monomial factor from a polynomial. (first introduced in Math 12) SK-FACTOR-GCF · Deepen
A	Determine the numerical and variable greatest common factor of polynomial terms. (first introduced in Math 12) SK-POLY-GCF · Apply
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M21-POL-004	I can factor a quadratic trinomial. TEXTBOOK SECTIONS 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials
SUPPORTING SKILLS	
A	Verify a factorization by multiplication. (first introduced in Math 12) SK-FACTOR-CHECK · Apply
R	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Reinforce
R	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Reinforce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

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SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M21-POL-005	I can factor a difference of squares and other required special cases. TEXTBOOK SECTIONS 6.3 Factor Special Products
SUPPORTING SKILLS	
A	Verify a factorization by multiplication. (first introduced in Math 12) SK-FACTOR-CHECK · Apply
I	Factor a difference of two squares. SK-FACTOR-DIFF-SQUARES · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M21-POL-006	I can use factoring to determine the zeros of a polynomial. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
A	Factor a difference of two squares. (first introduced in Math 21) SK-FACTOR-DIFF-SQUARES · Apply
A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
I	Apply the zero-product property. SK-FACTOR-ZERO-PRODUCT · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Rational Expressions and Equations

REQUIRED

UNIT TEXTBOOK COVERAGE

7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators; 7.5 Solve Rational Equations

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-RAT-001

I can state excluded values for a rational expression.
TEXTBOOK SECTIONS 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators

SUPPORTING SKILLS

I

Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · Introduce

M21-RAT-002

I can simplify a rational expression while preserving its restrictions.
TEXTBOOK SECTIONS 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

I

Cancel common factors rather than individual terms.
SK-RAT-CANCEL-FACTORS · Introduce

I

Factor numerators and denominators completely.
SK-RAT-FACTOR · Introduce

A

Determine excluded values from an original denominator. (first introduced in Math 21)
SK-RAT-RESTRICTIONS · Apply

M21-RAT-003

I can multiply and divide rational expressions.
TEXTBOOK SECTIONS 7.2 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

A

Add, subtract, multiply, and divide fractions. (inherited from middle school)
SK-NUM-FRACTION-OPERATE · Apply

A

Cancel common factors rather than individual terms. (first introduced in Math 21)
SK-RAT-CANCEL-FACTORS · Apply

A

Factor numerators and denominators completely. (first introduced in Math 21)
SK-RAT-FACTOR · Apply

A

Determine excluded values from an original denominator. (first introduced in Math 21)
SK-RAT-RESTRICTIONS · Apply

M21-RAT-004

I can add and subtract rational expressions using a common denominator.
TEXTBOOK SECTIONS 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators

SUPPORTING SKILLS

A

Add, subtract, multiply, and divide fractions. (inherited from middle school)
SK-NUM-FRACTION-OPERATE · Apply

I

Combine rational expressions over a common denominator.
SK-RAT-COMBINE · Introduce

I

Determine the least common denominator of rational expressions.
SK-RAT-LCD · Introduce

M21-RAT-005

I can solve a rational equation and reject extraneous solutions.
TEXTBOOK SECTIONS 7.5 Solve Rational Equations

SUPPORTING SKILLS

A

Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)
SK-ALG-INVERSE-OPERATIONS · Apply

I

Clear denominators from a rational equation.
SK-RAT-CLEAR-DENOMINATORS · Introduce

I

Check candidate solutions against denominator restrictions.
SK-RAT-EXTRANEIOUS · Introduce

A

Determine excluded values from an original denominator. (first introduced in Math 21)
SK-RAT-RESTRICTIONS · Apply

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SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Exponents and Radicals

REQUIRED

UNIT TEXTBOOK COVERAGE

8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.3 Simplify Rational Exponents; 8.4 Add, Subtract, and Multiply Radical Expressions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-EXP-001	I can simplify expressions containing negative exponents. TEXTBOOK SECTIONS 8.3 Simplify Rational Exponents
SUPPORTING SKILLS	
I	Rewrite a negative exponent using a reciprocal. SK-EXP-NEGATIVE · Introduce
A	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Apply
A	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Apply

M21-EXP-002	I can rewrite expressions between radical and rational-exponent forms. TEXTBOOK SECTIONS 8.3 Simplify Rational Exponents
SUPPORTING SKILLS	
I	Interpret numerator and denominator roles in a rational exponent. SK-EXP-RATIONAL · Introduce
I	Convert between radical and rational-exponent notation. SK-RAD-CONVERT · Introduce

M21-EXP-003	I can simplify radical expressions. TEXTBOOK SECTIONS 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.4 Add, Subtract, and Multiply Radical Expressions
SUPPORTING SKILLS	
A	Express a whole number as a product of prime factors. (inherited from middle school) SK-NUM-PRIME-FACTOR · Apply
I	Extract perfect-power factors from a radical. SK-RAD-FACTOR-PERFECT · Introduce

M21-EXP-004	I can apply exponent rules to expressions with integer and rational exponents. TEXTBOOK SECTIONS 8.3 Simplify Rational Exponents; 8.4 Add, Subtract, and Multiply Radical Expressions
SUPPORTING SKILLS	
A	Rewrite a negative exponent using a reciprocal. (first introduced in Math 21) SK-EXP-NEGATIVE · Apply
D	Apply the power rule for exponents. (first introduced in Math 12) SK-EXP-POWER · Deepen
D	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Deepen
D	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Deepen
A	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Apply
I	Apply the zero-exponent rule. SK-EXP-ZERO · Introduce

M21-EXP-005	I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Require an even-root radicand to be nonnegative when determining domain. SK-FUNC-DOMAIN-EVEN-ROOT · Introduce
I	Check a candidate solution for extraneous results. SK-RAD-CHECK · Introduce
I	Isolate a radical expression before solving. SK-RAD-ISOLATE · Introduce

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Quadratic Functions and Equations

REQUIRED

UNIT TEXTBOOK COVERAGE

9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form; 9.5 Solve Applications of Quadratic Equations; 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-QUA
D-001

I can identify key features of a quadratic function from its graph or equation.
TEXTBOOK SECTIONS 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

I	Identify standard, factored, and vertex forms. SK-QUAD-FORM-IDENTIFY · Introduce
I	Determine x- and y-intercepts of a quadratic function. SK-QUAD-INTERCEPTS · Introduce
I	Determine the vertex and axis of symmetry. SK-QUAD-VERTEX · Introduce

M21-QUA
D-002

I can connect standard, factored, and vertex forms to features of a parabola.
TEXTBOOK SECTIONS 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

I	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. SK-QUAD-FORM-CONVERT · Introduce
D	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Deepen
A	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Apply
A	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Apply
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M21-QUA
D-003

I can solve a quadratic equation by factoring.
TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
D	Apply the zero-product property. (first introduced in Math 21) SK-FACTOR-ZERO-PRODUCT · Deepen
I	Determine real and complex zeros using an appropriate method. SK-POLY-ZEROS · Introduce

M21-QUA
D-004

I can solve a quadratic equation using the square-root property.
TEXTBOOK SECTIONS 9.1 Solve Quadratic Equations Using the Square Root Property

SUPPORTING SKILLS

I	Apply the square-root property with both signs. SK-QUAD-SQUARE-ROOT · Introduce
A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply

M21-QUA
D-005

I can solve a quadratic equation by completing the square.
TEXTBOOK SECTIONS 9.2 Solve Quadratic Equations by Completing the Square

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Create a perfect-square trinomial by completing the square. SK-QUAD-COMPLETE-SQUARE · Introduce

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SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M21-QUA
D-006

I can solve a quadratic equation using the quadratic formula.
TEXTBOOK SECTIONS 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

I	Use the discriminant to predict the number and type of solutions. SK-QUAD-DISCRIMINANT · Introduce
I	Substitute coefficients accurately into the quadratic formula. SK-QUAD-FORMULA · Introduce
A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply

M21-QUA
D-007

I can choose an efficient method for solving a quadratic equation.
TEXTBOOK SECTIONS 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form

SUPPORTING SKILLS

A	Apply the zero-product property. (first introduced in Math 21) SK-FACTOR-ZERO-PRODUCT · Apply
A	Create a perfect-square trinomial by completing the square. (first introduced in Math 21) SK-QUAD-COMPLETE-SQUARE · Apply
A	Substitute coefficients accurately into the quadratic formula. (first introduced in Math 21) SK-QUAD-FORMULA · Apply
I	Select an efficient quadratic-solving method from equation structure. SK-QUAD-METHOD-SELECT · Introduce
A	Apply the square-root property with both signs. (first introduced in Math 21) SK-QUAD-SQUARE-ROOT · Apply

M21-QUA
D-008

I can graph a quadratic function using its vertex, intercepts, and symmetry.
TEXTBOOK SECTIONS 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

A	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. (first introduced in Math 21) SK-QUAD-FORM-CONVERT · Apply
D	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Deepen
D	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Deepen
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M21-QUA
D-009

I can create and interpret a quadratic model for a projectile, area, or optimization problem.
TEXTBOOK SECTIONS 9.5 Solve Applications of Quadratic Equations

SUPPORTING SKILLS

D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
I	Interpret vertex and zeros in an application. SK-QUAD-MODEL-FEATURES · Introduce
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

Math 21 In-Depth Curriculum

Algebra and its Functions

TEXTBOOK | OpenStax Intermediate Algebra 2e (Marecek and Mathis, 2020)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Extension Content

EXTENSION

UNIT TEXTBOOK COVERAGE

9.3 Solve Quadratic Equations Using the Quadratic Formula

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-EXT-001	I can perform basic arithmetic with complex numbers. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
I	Simplify integer powers of the imaginary unit. SK-CPLX-I-POWERS · Introduce
I	Add, subtract, and multiply complex numbers. SK-CPLX-OPERATE · Introduce

M21-EXT-002	I can express the nonreal solutions of a quadratic equation using complex numbers. TEXTBOOK SECTIONS 9.3 Solve Quadratic Equations Using the Quadratic Formula
SUPPORTING SKILLS	
A	Simplify integer powers of the imaginary unit. (first introduced in Math 21) SK-CPLX-I-POWERS · Apply
I	Express a quadratic solution with a negative discriminant in complex form. SK-CPLX-QUAD-SOLUTION · Introduce
A	Use the discriminant to predict the number and type of solutions. (first introduced in Math 21) SK-QUAD-DISCRIMINANT · Apply